

LESSON GUIDE

Technology Plus

Unit 1: Troubleshooting

1.4.1 – Troubleshooting Methodology

Lesson Overview:

In this lesson, students will identify the steps of the troubleshooting methodology and implement the methodology to solve similar problems a computing professional may encounter in the workforce.

Students will:

- Identify the steps of the troubleshooting methodology.
- Implement the troubleshooting methodology to model troubleshooting sample computer issues.

Guiding Question: How do computing professionals solve problems when they occur?

Suggested Grade Levels: 8-12

Technology Needed: Yes

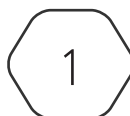
Approximate Time Required: 45 – 60 minutes

Standards:

CompTIA Tech+ FC0-U71 Objective

- 1.4 – Explain the troubleshooting methodology.
 - Identify the problem.
 - Establish a theory of probable cause (question the obvious).
 - Research knowledge base/Internet, if applicable
 - Test the theory to determine the cause.
 - Establish a plan of action to resolve the problem and implement the solution.
 - Verify full system functionality and, if applicable, implement preventive measures.
 - Document finding/lessons learned, actions, and outcomes.

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Lesson Background

Background Information:

Troubleshooting is one of the most practical skills students can develop, not just for IT careers but for problem-solving in everyday life. Many students come into a tech-focused class expecting clear-cut answers – press this button, follow that formula – but troubleshooting forces them to engage in critical thinking, persistence, and adaptability. In IT, even the most experienced professionals don't have every answer memorized; they rely on structured methods to break down problems and find solutions efficiently. By guiding students through this process, the lesson shifts their mindset from expecting instant fixes to understanding how systems work and why issues arise. This approach builds confidence, helping them see that mistakes and failures aren't roadblocks but essential steps toward mastering complex concepts. Plus, in a world where technology is constantly changing, the ability to troubleshoot effectively is far more valuable than memorizing specific technical solutions.

Materials Needed:

Materials per Class	• Lesson Guide • Lesson Slides 1.4.1 – Troubleshooting Methodology • Activity Slides 1.4.1 – Computer Not Starting • Activity 1.4.1 – Troubleshooting Scenarios (End of this document)
Materials per Student	• Student Notes 1.4.1 – Troubleshooting Methodology • Student Handout 1.4.1 – Troubleshooting Methodology • Assessment 1.4.1 – Troubleshooting Methodology (optional)

Cyber Terms:

Term	Definition
troubleshooting	the process of finding and fixing problems in a system, device, or software

1.4.1 – Troubleshooting Methodology

Guiding Question: How do computing professionals solve problems when they occur?

Lesson Launch (10 minutes)

Riddle Challenge: The Mystery Machine

- **Present this scenario:** A coffee machine in the break room isn't working. It won't power on, and no coffee is brewing. What's the first thing you do to solve the problem?
- **Goal:** Let students brainstorm their first steps. Guide them toward checking if the machine is plugged in, demonstrating the concept of "questioning the obvious."
- **Follow-up:** Discuss why this simple check is critical and how skipping it could waste time.

Logic Puzzle: The Faulty Lightbulb

- **Give this puzzle:** You have three switches outside a closed classroom, and only one of them controls the lightbulb inside. You can flip the switches as many times as you want, but you can only enter the room once to check the bulb. How do you figure out which switch controls the lightbulb?
- **Solution:** Flip one switch on for a few minutes, then turn it off. Flip a second switch on and enter the room. The bulb will either:
 - Be on (current switch is correct).
 - Be off but warm (first switch is correct).
 - Be off and cold (third switch is correct).
- **Goal:** Highlight systematic testing and observation, key components of troubleshooting.

Troubleshooting Methodology (15 minutes)

- Distribute Student Handout 1.4.1 – Troubleshooting Methodology and encourage them to fill in answers as you go through today's lesson.

Why Troubleshooting Matters

- Role in IT Careers.
 - **Troubleshooting** is a core skill for all IT roles, from help desk technicians to cybersecurity analysts.
 - Example: If you've ever called tech support or asked someone why the Wi-Fi isn't working, you've relied on someone using troubleshooting skills.
 - Emphasize that troubleshooting isn't about memorizing fixes – it's about understanding systems

and solving problems logically.

Efficiency and Impact

- Good troubleshooting saves time and resources.
- Example: Imagine a company's network goes down. Every minute the problem goes unresolved, it costs money and impacts employees, customers, and reputation

Transferable Skills

- Troubleshooting teaches critical thinking, persistence, and patience, which are useful beyond IT.
- Example: Figuring out why a car won't start or why your phone won't charge also uses troubleshooting.

Overview of Methodology

- Troubleshooting methodology will feel very similar to the scientific method or the engineering design process to students. Troubleshooting does involve a lot of the same processes from methods students have learned about in the past.
 - Walk through each step briefly, using a relatable example like a computer not connecting to Wi-Fi.
 - Use interactive check-ins after introducing each by asking students what they might do at each step of the process.
1. Identify the Problem.
 - Understand the symptoms.
 - Example: Someone says, "My laptop can't connect to the Wi-Fi." Ask questions: Did this just start? Do other devices work?
 - Watch your tone when asking questions. You don't want your tone to be accusatory. Prematurely assigning blame for the problem won't help you reach a solution any faster.
 2. Establish a Theory of Probable Cause.
 - Question the obvious and make educated guesses.
 - Example: Is the Wi-Fi on? Did the password change?
 - Research Knowledge Base/Internet (if applicable).
 - Look up similar issues or error codes.
 - Example: Search for error messages online to see if it's a common problem and has solutions.

3. Test the Theory to Determine the Cause.
 - o Test one possible cause at a time.
 - o Check simple stuff first.
 - o Example: Is Wi-Fi turned on for the laptop? Is the laptop in Airplane Mode?
 - o Check to see whether it's a user error.
 - o Example: If you think the issue is a bad password, re-enter to confirm it.
 - o Restart the computer – you'd be amazed at how often a simple reboot can solve a problem.
4. Establish a Plan of Action and Implement the Solution.
 - o Plan how to fix the issue with minimal disruption.
 - o Example: If the password needs to be reset, you might need to coordinate with someone who has the authority to do that.
5. Verify Full System Functionality and Implement Preventive Measures.
 - o Test to make sure the fix worked without causing problems to another part of the system.
 - o If possible, implement preventative measures.
 - o Example: Make sure the laptop connects and recommend that the user save the new password.
 - o If the preventive measures you implement require education on changing the way a user does something, remember to be kind. Things that seem obvious to you may not be obvious to the person you are helping.
6. Document Findings, Actions, and Outcomes.
 - o Chances are you won't remember how you solved a particular problem a month after you've solved it.
 - o Document what the problem was, how it was fixed, and what to do next time.
 - o Carrying a personal notebook to take notes on how you solved a problem is a great idea.
 - o Some organizations have logs or ticket systems where you can detail how you solved a problem so that everyone who has access to the log or system can gain insight on how to solve the problem.
 - o There are several forms of system-based documentation out there. The most important part of any system is that you use it!

Activity: Scenario – Computer Not Starting (5-10 minutes)

- Use this scenario to lead the whole class through the steps of the troubleshooting methodology.
- On the testing theories slide there are several options that link to other places in the presentation choose-your-own-adventure-style.
- Each theory tested that doesn't fix the problem explains why the theory was plausible but leads back to the theories page.
- The "Try a Different Outlet" theory is the correct answer, and its link will move you forward to the next step in the methodology.
- Discuss the answers to the questions on the slide.

Activity: Troubleshooting Scenarios (10-15 minutes)

- Print and cut up the Activity 1.4.1 – Troubleshooting Scenarios handouts located at the end of this document before class begins.
- Group students and give each group a scenario.
- Have the group complete the Troubleshooting Scenarios Report part of the Student Handout as they brainstorm their problem.
- Once students have developed their list of theories, give students the actual solution for the scenario so that they can compare their theories to the answer and complete their report.
- After students have finished their reports, invite each group to present their scenario and report to the class.

Putting It All Together (5-10 minutes)

- What was the most impactful thing you learned during today's lesson?

Student answers will vary.

- Explain that troubleshooting involves curiosity, patience, and teamwork.

Activity: Scenario – Troubleshooting Scenarios to Cut Out

- Print this page and cut the scenarios out to distribute to students during the activity. Retain the solutions to give to students after they have developed their lists of theories.

Keyboard isn't working
Application crashes on startup
Laptop won't connect to Wi-Fi
The printer won't print
Monitor displays a black screen
Slow computer performance
External drive not recognized
No sound from speakers
Software installation fails
Website not loading

Solutions

- Keyboard isn't working: The keyboard's USB connection is loose. Reconnect it securely or replace the batteries if wireless.
- Application crashes on startup: The application is outdated and incompatible with the operating system. Update application to the latest version.
- Laptop won't connect to Wi-Fi: The router password was recently changed. Update the saved Wi-Fi password on the laptop.
- The printer won't print: The printer was paused in the print queue. Resume the printer in the settings.
- Monitor displays a black screen: The monitor's input was set to VGA, but the connected cable was HDMI. Change the input source to HDMI.
- Slow computer performance: Multiple background applications were running. Open Task Manager and close unnecessary programs.
- External drive not recognized: The drive is not formatted for the operating system. Format the drive or use a compatible file system.
- No sound from speakers: The computer's default audio was set to headphones instead of speakers. Change the default device in the audio settings.
- Software installation fails: There wasn't enough disk space. Free up space by deleting unnecessary files or moving them to external storage.
- Website not loading: The website is down for maintenance. Confirm this using an online status checker or wait and try again later.